CHAPTER NINTEEN: THE PROCESS MODEL OF MANAGEMENT

The Process Model: A Systematic Approach to Quality in everything we do.

Modern organizations operate in an environment of complexity, competition, and constant change. ISO 9001, the internationally recognized Quality Management System (QMS) standard, addresses this reality by promoting a process-based approach to managing organizations. At the heart of this approach lies the Process Model, supported by the PDCA (Plan–Do–Check–Act) cycle, which enables organizations to deliver consistent results while continually improving performance.

UNDERSTANDING THE PROCESS-BASED APPROACH

A **process** is a set of interrelated or interacting activities that transform inputs into outputs. ISO 9001 encourages organizations to view their operations not as isolated functions or departments, but as connected processes that collectively create value for customers.

Instead of focusing on individual tasks, the process model asks key questions:

- What inputs are required?
- What activities transform those inputs?
- What outputs are produced?
- Who is responsible?
- How is performance measured?

- What risks and opportunities exist?

For example, customer requirements enter the organization as inputs. These inputs pass through multiple processes—design, procurement, production, delivery, and support—and emerge as products or services that must meet customer expectations. Feedback from customers then becomes an input for improvement.

This end-to-end view ensures alignment, accountability, and efficiency.

Core Elements of the ISO 9001 Process Model

The ISO 9001 process model integrates several essential elements:

1. **Customer Focus:** Customer needs and expectations initiate the process chain and define success. Satisfaction is not assumed—it is measured and acted upon.
2. **Leadership and Context:** Top management establishes purpose, direction, and priorities. Understanding the organization's internal and external context ensures processes remain relevant.
3. **Process Interaction:** Outputs of one process often become inputs to another. Weakness in one process can disrupt the entire system, making coordination critical.
4. **Performance Measurement:** Each process must have defined objectives and indicators to assess effectiveness and efficiency.
5. **Continual Improvement:** Improvement is not a one-time effort but a permanent organizational mindset.

THE PDCA CYCLE: ENGINE OF THE PROCESS MODEL

The PDCA cycle, also known as the Deming Cycle, is the operational backbone of ISO 9001. It provides a structured method for managing processes and driving improvement.

1. Plan

The Plan phase establishes the foundation for effective execution.

Key activities include:

- Understanding customer requirements and organizational objectives
- Defining processes and responsibilities
- Identifying risks and opportunities
- Setting quality objectives and performance indicators
- Planning resources, timelines, and controls

In ISO 9001, planning is closely linked to risk-based thinking. Organizations must anticipate what could go wrong, what could go right, and how both should be addressed.

Example:

A manufacturing company plans a new product line by defining specifications, supplier criteria, quality standards, and inspection methods before production begins.

2. Do

The **Do** phase involves implementing what was planned.

Key activities include:

- Operating processes as defined
- Training personnel
- Producing products or delivering services
- Maintaining records and documentation

Consistency is crucial at this stage. Well-defined processes reduce variation, errors, and dependency on individuals.

Example:

The manufacturing team produces goods according to approved work instructions, while procurement follows defined supplier evaluation procedures.

3. Check

The Check phase focuses on monitoring and evaluating performance.

Key activities include:

- Measuring process outputs against objectives
- Conducting internal audits
- Reviewing customer feedback and complaints
- Analysing data and trends

This phase answers a critical question: Are we achieving what we planned?

ISO 9001 emphasizes evidence-based decision-making. Opinions and assumptions are replaced with data, metrics, and documented findings.

Example:

The company analyses defect rates, delivery delays, and customer complaints to assess whether quality targets are being met.

4. Act

The **Act** phase converts learning into improvement.

Key activities include:

- Taking corrective actions for nonconformities
- Preventing recurrence of problems
- Improving processes, controls, or objectives

- Updating documentation and training

This phase ensures that mistakes are not repeated and that successes are standardized.

Example:

If frequent defects are traced to a supplier, the organization may revise supplier selection criteria, improve inspections, or switch vendors altogether.

PDCA as a Continuous Loop

PDCA is cyclical. The "Act" phase feeds into the next "Plan" phase, creating a loop of continuous improvement.

Over time, this leads to:

- Reduced waste and rework
- Higher customer satisfaction
- Better risk management
- Stronger organizational learning

ISO 9001 embeds PDCA at multiple levels:

- Across the entire Quality Management System
- Within individual processes
- For improvement initiatives and corrective actions

Benefits of the ISO 9001 Process Model

Organizations that effectively implement the process model experience several advantages:

- Consistency in product and service quality
- Transparency in roles, responsibilities, and performance
- Improved efficiency through clear workflows
- Better decision-making based on data
- Stronger customer trust and credibility
- Scalability, making growth manageable

For startups and growing businesses, the process model prevents chaos. For mature organizations, it ensures discipline and adaptability.

RISK-BASED THINKING

Every organization operates in an environment of uncertainty. Markets shift, suppliers fail, technologies evolve, regulations change, and customer expectations rise. The risk-based model, as emphasized in ISO 9001, provides a structured approach to anticipating uncertainty, preventing failures, and improving outcomes. Rather than reacting to problems after they occur, risk-based thinking encourages organizations to identify, assess, and address risks before they impact performance.

ISO 9001 does not require a formal risk management system, but it embeds risk-based thinking throughout the Quality Management System (QMS). The intent is simple yet powerful:

- Identify what could go wrong
- Understand what could go right
- Take proportionate actions

Risk-based thinking replaces checklist compliance with judgment, foresight, and accountability. It applies to strategic decisions, operational processes, supplier management, customer satisfaction, and continual improvement.

Understanding Risk and Opportunity

- **Risk** is the effect of uncertainty on objectives (often negative)

- **Opportunity** is the positive side of uncertainty

Both must be addressed together. Avoiding risk without recognizing opportunity can lead to stagnation, while chasing opportunity without understanding risk can lead to failure.

Example:

Launching a new product involves risks (technical failure, low demand) but also opportunities (market leadership, revenue growth).

The Risk-Based Model: A Structured Approach

The risk-based model typically follows four logical stages:

1. Risk Identification
2. Risk Assessment
3. Risk Prioritization
4. Risk Treatment and Review

This model integrates seamlessly with the PDCA cycle, ensuring risks are continuously reviewed and improved.

1. Risk Identification: Knowing What Can Go Wrong

Risk identification is the foundation of effective risk management. At this stage, organizations systematically identify risks that may affect objectives, processes, or compliance.

Sources of risk include:

- Internal factors: skills gaps, process failures, equipment breakdowns
- External factors: market changes, supplier reliability, regulations
- Strategic risks: competition, technology disruption
- Operational risks: quality defects, delivery delays
- Compliance risks: legal and regulatory non-conformance

Tools commonly used:

- Process mapping
- Brainstorming and expert judgment
- SWOT analysis
- Past incidents and audit findings
- Customer complaints and feedback

Example:

A service company identifies risks such as employee attrition, data breaches, and inconsistent service quality.

2. Risk Assessment: Understanding Impact and Likelihood

Once risks are identified, they must be **assessed** to understand their seriousness. Risk assessment evaluates two key dimensions:

Likelihood

How probable is the risk event?

- Rare
- Possible
- Likely

Impact

What would be the consequence if it occurs?

- Minor inconvenience
- Financial loss
- Reputational damage
- Legal consequences

Some organizations also consider **detectability**—how easily the risk can be identified before causing harm.

Risk assessment may be:

- **Qualitative** (low, medium, high)
- **Semi-quantitative** (scored scales)
- **Quantitative** (financial or statistical models)

Example:

A supplier delay might be “likely” but with “moderate” impact, while a data breach may be “unlikely” but with “severe” impact.

3. Risk Prioritization: Focus Where It Matters Most

Not all risks deserve equal attention. **Risk prioritization** ensures that resources are directed to the most significant risks.

This is often achieved by combining likelihood and impact to determine:

- High priority risks (immediate action required)
- Medium priority risks (monitor and control)
- Low priority risks (accept or review periodically)

In more structured systems, organizations use:

- Risk matrices
- Risk priority numbers (RPNs)
- Heat maps

The goal is practical decision-making, not paperwork.

Example:

A startup may prioritize cash-flow risk over minor process inefficiencies, as survival depends on liquidity.

4. Risk Treatment: Deciding What to Do About Risk

Once risks are prioritized, organizations choose appropriate responses. Common strategies include:

- **Avoid**: Stop the activity causing the risk
- **Reduce**: Implement controls to lower likelihood or impact
- **Transfer**: Shift risk through insurance or outsourcing
- **Accept**: Acknowledge the risk when cost of control outweighs benefit

ISO 9001 emphasizes **proportionate action**—controls should be appropriate to the level of risk.

Example:

A company reduces quality risk by adding inspections, while transferring financial risk through insurance.

Risk-Based Thinking Across the Organization

Risk-based thinking is not limited to top management. It applies at all levels:

- **Strategic level**: Market entry, investments, expansion
- **Process level**: Production, service delivery, procurement
- **People level**: Training, competence, succession planning
- **Customer level**: Expectations, complaints, satisfaction

Embedding risk awareness into daily decisions creates a culture of prevention rather than correction.

Monitoring and Review: Keeping Risk Management Alive

Risks are not static. Changes in technology, regulation, or market conditions can alter risk profiles.

Effective organizations:

- Review risks periodically
- Monitor effectiveness of controls
- Learn from incidents and near misses
- Update risk registers and action plans

This aligns with the **Check and Act** stages of PDCA, reinforcing continual improvement.

Benefits of a Risk-Based Model

Organizations that adopt risk-based thinking experience:

- Fewer surprises and disruptions
- Improved compliance and audit outcomes

- Better decision-making under uncertainty
- Increased customer confidence
- Enhanced resilience and sustainability

For entrepreneurs, the risk-based model balances ambition with realism. For mature organizations, it ensures stability without complacency.

The risk-based model transforms uncertainty from a threat into a managed variable. By systematically identifying risks, assessing their significance, prioritizing action, and reviewing outcomes, organizations move from reactive firefighting to proactive governance.

Aligned with ISO 9001, risk-based thinking is not about eliminating risk—it is about understanding it, managing it intelligently, and using it to create better, more reliable outcomes.

CHAPTER TWENTY: IDEA TO SCALABLE OPERATIONS

In the early stages of entrepreneurship, clarity is elusive and resources are limited. The Minimum Viable Product (MVP) is your strategic weapon—it's the leanest version of your product that delivers core value to early adopters while enabling rapid learning. It's not about building something perfect; it's about building something purposeful.

We will walk through the full lifecycle from ideation to MVP launch, covering product development frameworks, validation techniques, testing protocols etc.

Identifying the Problem that is sitting with a reward:

Every great product starts with a problem worth solving. Ask:

- Who is experiencing pain? Is that causing loss?
- How frequent and intense is the pain?
- What existing solutions fall short?
- Is the customer willing to pay for you to find a way to get rid of the pain. If yes, how much?

Use tools like Customer interviews, Observation, Online forums and reviews

WHAT IS AN MVP?

An MVP, the minimum viable product is the simplest version of your product that:

- Solves a core problem
- Delivers real value

- Enables feedback and learning

It's not a prototype or beta—it's a functional product with minimal features.

Pay attention to the various dimensions of viability. It is not a Minimum viable Idea. It has to be a product that works to deliver value to customer.

MVP VS. MLP VS. MAP

Term	Description
MVP (Minimum Viable Product)	Core functionality for early adopters
MLP (Minimum Lovable Product)	Adds delight and emotional appeal
MAP (Minimum Awesome Product)	Combines utility and wow factor

Start with MVP, evolve toward MLP and MAP.

THE FOUR LAYERS:

Entrepreneurship and innovation often begin with a scientific or technological insight, but the journey from concept to marketable product involves several crucial stages. These can be broadly understood as four interconnected pillars: Fundamental Science, Application Development, Digitisation, and Market Expansion. Each stage adds value, refines the idea, and increases the likelihood of commercial success.

1. Fundamental Science

The journey begins with fundamental science—the basic understanding of principles, mechanisms, or phenomena. This is

the intellectual foundation that informs all subsequent innovation.

- **Purpose**: Fundamental science seeks to answer the "why" and "how" behind natural or technological processes.
- **Scope**: It may involve experiments, theoretical models, simulations, or data analysis.
- **Relevance**: Not all scientific discoveries are immediately commercially viable, but they provide **knowledge capital** that can lead to breakthroughs in the future.

Example: The discovery of semiconductors' electrical properties was pure science initially, with no immediate consumer application. However, this foundational knowledge eventually enabled the development of microchips, smartphones, and modern computing.

For entrepreneurs, understanding the scientific principles behind a product or service ensures credibility, reduces technical risk, and enables informed decision-making during development.

2. Application Development

Once a scientific concept is understood, the next step is application development—translating theory into something practically useful in the current context.

- **Focus**: Identify whether the science can solve a real-world problem at this time and in the relevant market.
- **Process**: Develop prototypes, test use cases, and iterate on the design.
- **Considerations**: Technical feasibility, scalability, cost, safety, and regulatory requirements must all be evaluated.

Example: CRISPR gene-editing technology originated as a scientific discovery. Application development turned it into therapies for genetic disorders, agricultural improvements, and diagnostic tools.

For entrepreneurs, this stage requires bridging the gap between theory and practice. It often involves collaboration between scientists, engineers, designers, and industry experts to ensure that the technology is both functional and market-ready.

3. Digitisation

Digitisation is the process of leveraging technology to make the application more efficient, accessible, and affordable. It enhances scalability and user adoption.

- **Purpose**: Digitisation transforms a physical or labour-intensive process into a technology-driven solution.
- **Benefits**:
    - Reduces operational costs and manual effort
    - Improves speed, accuracy, and convenience
    - Expands reach to new customers through digital platforms
- **Tools**: Software, cloud computing, mobile applications, data analytics, and AI/ML are commonly used to digitise products and services.

Example: Telemedicine platforms digitise healthcare delivery, enabling remote consultations, electronic prescriptions, and patient record management. What was once limited to physical clinics becomes accessible, affordable, and scalable.

For startups, digitisation can be the difference between a niche prototype and a mass-market solution. It often opens opportunities for subscription models, recurring revenue, and broader social impact.

4. Expand and Market

The final stage is **expansion and market deployment**—bringing the product or service to a **wider audience** and creating sustainable impact.

- **Strategy**: Entrepreneurs must define target customers, distribution channels, pricing, and promotion.
- **Scaling**: Efficient production, supply chain management, and customer support systems are critical.
- **Market Feedback**: Continuous iteration based on user feedback ensures product-market fit and long-term success.

Example: Electric vehicles (EVs) started with scientific research on battery chemistry, moved through prototype development, became digitised through smart vehicle software and mobile apps, and are now expanding rapidly into global markets with charging infrastructure, financing options, and marketing campaigns.

This stage requires marketing acumen, operational excellence, and strategic partnerships to ensure that the innovation does not remain a niche or pilot but achieves widespread adoption and impact.

Integrating the Four Pillars

- **Sequential yet iterative**: While presented as a sequence, these pillars often **overlap**. Insights from market feedback can inform further application development or inspire new scientific research.
- **Risk mitigation**: Following this structured approach reduces technical, operational, and market risks.

- **Value creation**: Each stage progressively increases the utility, accessibility, and profitability of the innovation.

In essence, the four pillars—Fundamental Science, Application Development, Digitisation, and Market Expansion—form a framework for translating knowledge into actionable, scalable solutions. Entrepreneurs who understand and implement these stages systematically are more likely to convert ideas into successful, impactful ventures.

MVP TESTING PROTOCOLS

Alpha Testing

- Internal team or close network
- Focus on bugs, usability, and flow
- Use feedback forms and screen recordings

Beta Testing

- External early adopters
- Focus on real-world usage
- Offer incentives for feedback

Metrics to Track

Metric	Why It Matters
Activation Rate	% of users who complete key action
Retention Rate	% of users who return over time
Churn Rate	% of users who stop using the product
Feature Usage	Identifies what's working

FEEDBACK LOOPS AND ITERATION

Build–Measure–Learn Loop

1. Build MVP based on hypotheses

2. Measure user behaviour and feedback
3. Learn what works and what doesn't
4. Iterate rapidly

Feedback Channels

- In-app surveys
- Customer interviews
- Support tickets
- Social media listening

Soft Launch vs. Hard Launch

Launch Type	Description	Use Case
Soft Launch	Limited release, controlled feedback	Ideal for MVP
Hard Launch	Full-scale public release	After validation and iteration

Start small, learn fast, scale wisely.

Go-to-Market (GTM) Plan

- Define target audience
- Choose launch channels (email, social, PR)
- Prepare onboarding and support
- Monitor metrics in real-time

FEATURE PRIORITIZATION

Use frameworks like RICE or Kano Model. There are many more models that could be used.

- RICE: Reach, Impact, Confidence, Effort - The RICE model is a practical framework used to prioritize product features and initiatives based on four key dimensions: Reach, Impact, Confidence, and Effort. It helps teams make data-informed decisions by quantifying the

potential value of each idea. Reach estimates how many users or customers will be affected by the feature within a given time frame. Impact assesses how strongly the feature will influence user behaviour or business goals—typically rated on a scale (e.g., minimal to massive). Confidence reflects how certain you are about your reach and impact estimates, helping to account for assumptions and unknowns. Finally, Effort measures the time and resources required to implement the feature, usually in person-hours or weeks. The RICE score is calculated as (Reach × Impact × Confidence) ÷ Effort, allowing entrepreneurs to prioritize high-value, low-effort features that can accelerate learning and growth—especially critical when building a Minimum Viable Product (MVP) under resource constraints.

- Kano Model: Basic, Performance, Delight features - The Kano Model is a powerful framework for prioritizing product features based on how they impact customer satisfaction. Developed by Professor Noriaki Kano, it categorizes features into three main types: Basic Needs, which customers expect and take for granted (their absence causes dissatisfaction but their presence doesn't delight); Performance Needs, which directly correlate with satisfaction—the better you deliver, the happier the customer; and Delighters, which are unexpected features that pleasantly surprise users and create emotional engagement. By mapping features across these categories, entrepreneurs can make strategic decisions about what to include in their Minimum Viable Product (MVP), ensuring that limited resources are focused on elements

that drive both utility and user delight. This model is especially useful in iterative development, helping teams balance functionality with innovation.

Scaling Infrastructure

- Move from shared hosting to cloud (AWS, Azure)
- Implement CI/CD pipelines
- Optimize database queries

Team Expansion

- Hire product managers, QA engineers, customer success reps
- Build cross-functional squads
- Invest in culture and documentation

Common Pitfalls and How to Avoid Them

Pitfall	Impact	Solution
Overbuilding	Wastes time and money	Focus on core value
Ignoring feedback	Misses market signals	Build feedback loops
Premature scaling	Breaks operations	Validate before scaling
Poor testing	Damages user trust	Invest in QA
Lack of documentation	Slows team onboarding	Create living product docs

QUALITY MANAGEMENT

Quality is one of the most critical determinants of success for any organization. Whether an organization offers tangible products or intangible services, quality strongly influences customer satisfaction, reputation, profitability, and long-term survival. In highly competitive markets, quality is no longer a differentiator alone—it is a basic expectation. Customers assume that products and services will perform reliably, safely, and consistently.

We now will see the importance of quality in products and services, highlighting its impact on customers, organizations, and society, and explaining why quality must be treated as a strategic priority.

Understanding Quality in Products and Services

Quality can be broadly defined as the ability of a product or service to meet or exceed customer expectations. It includes aspects such as performance, reliability, durability, safety, and consistency.

For products, quality is often visible and measurable—for example, size, strength, finish, or life of the product. For services, quality is more subjective and is experienced during service delivery, such as responsiveness, courtesy, accuracy, and timeliness.

High-quality products and services satisfy customer needs and expectations. When customers receive what they expect—or more—they feel satisfied and confident in their purchase decision. This is what is called Customer Satisfaction

Satisfied customers are more likely to:

* Repeat purchases
* Recommend the brand to others
* Remain loyal even when competitors offer alternatives

Customer Trust and Loyalty

Consistent quality builds trust. Customers prefer brands they can rely on without risk. Over time, trust develops into loyalty, reducing the likelihood of customers switching to competitors.

In services such as healthcare, banking, education, and transportation, trust based on quality is especially critical.

Competitive Advantage

Quality provides a strong competitive advantage. Organizations known for quality often command higher market share and, in some cases, premium prices.

In global markets, quality is often a prerequisite for entry. Poor quality can quickly eliminate an organization from competition.

Contrary to the belief that quality is expensive, poor quality actually costs more. Defects, rework, returns, complaints, and warranty claims increase operating costs.

Good quality reduces, Scrap and rework, Customer complaints, Warranty and service costs

This concept is often referred to as the cost of poor quality.

Improved Productivity and Efficiency: Well-designed processes that focus on quality reduce errors and interruptions. Employees spend less time correcting mistakes and more time adding value.

As a result, productivity improves and processes become more efficient.

Importance of Quality in Services - Service Experience and Perception

Since services are produced and consumed simultaneously, quality is judged largely by customer experience. Factors such as employee behavior, speed of service, and problem resolution strongly influence customer perception.

Even a technically correct service can be perceived as poor if delivered rudely or late.

Service industries such as hospitality, airlines, healthcare, and IT depend heavily on repeat customers. High service quality increases customer retention, which is often more cost-effective than acquiring new customers.

Quality and Brand Image - Reputation and Goodwill

Quality shapes an organization's reputation. A single quality failure can damage brand image, while consistent quality builds goodwill over time.

In the age of social media and online reviews, poor quality can spread rapidly and affect public perception.

Long-Term Brand Value: Strong brands are built on consistent quality. Customers associate certain brands with reliability, safety, or superior performance, giving the organization long-term advantages.

Quality is Profitable: High quality leads to higher customer satisfaction, increased sales, and reduced costs—all of which improve profitability.

Organizations that invest in prevention and continuous improvement often achieve better financial performance than those that focus only on inspection.

Quality products and services enable organizations to enter new markets, meet international standards and win large and long-term contracts

Safety and Consumer Protection: Quality is directly linked to safety. Poor-quality products can cause accidents, health hazards, and environmental damage.

Organizations have a responsibility to ensure that their products and services are safe and reliable.

Sustainability and Ethics: Quality also contributes to sustainability by reducing waste, conserving resources, and promoting responsible production and consumption.

Ethical business practices require organizations to deliver what they promise in terms of quality.

Quality can act as a Strategic Tool and a differentiator. Quality should not be treated as an operational issue alone. It is a strategic tool that influences long-term success.

Organizations that embed quality into their culture and strategy are better equipped to adapt to changing customer needs and market conditions.

Role of Management: Top management plays a crucial role in promoting quality by setting clear quality objectives, providing resources and training and encouraging continuous improvement

Without leadership commitment, quality initiatives are unlikely to succeed.

EVOLUTION OF QUALITY MANAGEMENT

Quality management did not appear overnight as a neat set of standards or certifications. It evolved slowly, shaped by industrial revolutions, wars, competition, customer expectations, and hard-earned managerial lessons. What began as simple inspection at the end of production has today become a strategic, organization-wide philosophy focused on customer value, continuous improvement, and sustainability. This chapter traces the journey of quality management across time, highlighting key ideas, thinkers, and practices that transformed quality from a technical function into a core management discipline.

Quality in the Pre-Industrial Era: Craftsmanship and Pride: Before factories came in, quality was inseparable from craftsmanship. Artisans and craftsmen personally designed, produced, and inspected their work. The same individual was responsible for raw material selection, production, and final output. The same was true for services as well.

Key characteristics of this era included:

- Individual responsibility for quality - the craftsman's reputation depended on the quality of work.
- Customization over standardization– products were made to order.
- Implicit quality standard – based on skill, tradition, and guild rules rather than written specifications.
- Quality was ensured through skill and pride rather than systems. However, this approach could not scale when demand increased.

Industrial Revolution: Inspection-Based Quality: The Industrial Revolution brought mass production, division of labor, and

machinery. While productivity increased dramatically, quality suffered because workers no longer owned the entire product. They generally were not even aware of how their part of the work impacted the final use at the customer's end.

To address this, organizations introduced inspection as a separate function. Inspectors were like customer's representative. It was not abnormal to find customers posting their inspectors on the assembly line.

Features of this stage:

Quality was checked after production
Inspectors identified defective items.
Defects were accepted as inevitable.
Focus was on sorting good products from bad ones.

This approach treated quality as a 'detection problem', not a prevention problem. Costs of rework and scrap were high, but management largely accepted them as part of doing business.

One can compare this stage with the way agricultural produce is inspected, sorted and graded. From the same mango tree, several grades of mangoes were produced.

Early 20th Century: Statistical Quality Control

As production became mechanical and as volumes grew, inspection alone became inefficient and expensive. This led to the application of statistics to quality problems. Process control on machines were far easier than process control by hand.

In the 1920s, Statistical Process Control (SPC) came up. This concept distinguished between:

Common causes of variation (inherent in the process) Vs.
Special causes of variation (due to assignable factors)

Control charts enabled organizations to:

Monitor processes during production
Detect abnormal variations early
Control quality rather than merely inspect it

This marked a fundamental shift: Quality moved from product inspection to process control.

World War II: Quality as a National Priority

During World War II, the need for reliable weapons, ammunition, and equipment made quality critical. Governments demanded consistency and reliability at massive scale.

Key developments:

Large-scale use of SPC in manufacturing
Standardized inspection procedures
Training programs for quality engineers

After the war, many Western companies abandoned these rigorous practices. Ironically, Japan adopted and refined them.

Post-War Japan: The Quality Revolution

Japan, rebuilding its economy, embraced quality as a survival strategy. American quality experts were invited to train Japanese engineers and managers.

Key Contributors

W. Edwards Deming

Emphasized management responsibility for quality
Introduced the “Plan–Do–Check–Act (PDCA) cycle”
Advocated continuous improvement

Joseph M. Juran

Defined quality as “fitness for use”
Introduced the Juran Trilogy - Planning, Control, Improvement

Kaoru Ishikawa

Developed the “cause-and-effect (fishbone) diagram”
Promoted company-wide quality control
Encouraged employee involvement through quality circles

Japanese companies embedded quality into culture, leading to global competitiveness in automobiles and electronics.

Quality Assurance: System-Based Thinking

By the 1960s and 1970s, quality thinking expanded beyond production to include design, procurement, and documentation. This stage is known as Quality Assurance (QA).

Characteristics:

Prevention rather than detection
Documented procedures and work instructions
Design reviews and process audits
Supplier quality assurance

Quality became a ‘system’, not a department. This laid the foundation for international quality standards.

Total Quality Management (TQM): Organization-Wide Quality

In the 1980s, faced with intense global competition, Western companies rediscovered quality. This gave rise to Total Quality Management (TQM).

TQM principles include:

Customer focus
Leadership commitment
Employee involvement
Continuous improvement
Data-based decision-making

Quality was no longer confined to manufacturing. It extended to:

Marketing
Finance
Human resources
Customer service

TQM transformed quality into a management philosophy.

Standardization and ISO 9000 Series

To create common benchmarks, the International Organization for Standardization (ISO) introduced the ISO 9000 series in 1987.

Key features:
Emphasis on documented processes
Consistency in operations
Customer satisfaction
External certification

ISO standards helped organizations to:

Demonstrate credibility
Improve process discipline
Compete in global markets

However, critics noted that certification sometimes became a paperwork exercise rather than true quality improvement.

Six Sigma: Defect Reduction and Financial Impact

In the 1990s, Motorola and later General Electric popularized "Six Sigma".

Core ideas:

* Focus on reducing variation
* Target of 3.4 defects per million opportunities
* Structured problem-solving
* Strong linkage between quality and financial performance

Six Sigma elevated quality from an operational issue to a strategic business initiative.

Lean and Integrated Quality Approaches

Lean manufacturing, originating from the Toyota Production System, emphasized:

* Waste elimination
* Flow efficiency
* Value from the customer's perspective

When combined with Six Sigma, organizations adopted Lean Six Sigma, integrating speed and quality.

Quality management now addressed:

Cost
Delivery
Flexibility
Customer experience

Modern Quality Management: Digital and Customer-Centric

Today, quality management operates in a digital, service-oriented, and sustainability-focused environment.

Current trends include:

Real-time quality analytics
Automation and AI-driven inspection
Quality in services and software
Customer experience management
ESG and sustainability integration

With the social media explosion, Quality is also seen as perceived value.

The evolution of quality management reflects a profound shift in managerial thinking. From inspection to prevention, from products to processes, and from departments to organizational culture, quality has become a strategic imperative.

Organizations that understand this evolution recognize a simple truth: Quality is not a cost to be minimized but an investment that drives long-term success.

CHAPTER TWENTY ONE: MANAGEMENT PROCESSES

Operations management is the silent engine behind every successful business. While strategy sets the direction and marketing drives demand, operations ensure that products and services are delivered consistently, efficiently, and with quality. For entrepreneurs, mastering operations means transforming vision into execution—day after day, customer after customer.

We explore, the core pillars of operations management: process design, supply chain optimization, quality control, and customer support. Operational excellence is your foundation for scale and Operations planning has to be fully integrated with product development and business planning.

DESIGNING EFFICIENT AND REPEATABLE PROCESSES

A process is a set of interrelated or interacting activities that transforms inputs into outputs using resources, under defined conditions, to achieve intended results. In modern management systems—especially ISO 9001 and risk-based frameworks—organisations are viewed not as departments, but as networks of processes.

Effective process definition ensures consistency, predictability, accountability, and continuous improvement.

1. Super Processes (Macro Processes)

Super processes (also called core, macro, or value-chain processes) represent the highest level of process classification. They cut across departments and directly contribute to the organisation's strategic objectives and customer value.

Characteristics

- End-to-end in nature
- Cross-functional
- Directly linked to customer satisfaction and business outcomes
- Usually owned by senior leadership

Typical Super Processes

- Order to Cash (Customer inquiry → delivery → payment)
- Concept to Market (Idea → design → production → launch)
- Procure to Pay (Supplier selection → purchase → payment)
- Hire to Retire (Recruitment → employment → exit)

Purpose

- Align operations with strategy
- Enable top-level monitoring and governance
- Provide a framework for breaking down complexity

2. Sub Processes (Operational and Support Processes)

Definition

Sub processes are components of super processes. They perform specific, well-defined functions and are usually owned by functional managers.

Characteristics

- Narrower in scope

- Clearly defined boundaries
- Easier to measure and control
- Often documented in SOPs or work instructions

Example

Super Process: Order to Cash

Sub Processes:

- Order review and confirmation
- Credit check
- Production planning
- Dispatch and logistics
- Invoicing and collections

Importance

- Enable operational control
- Clarify roles and responsibilities
- Support training and standardisation

3. Inputs

Inputs are the elements required to initiate and execute a process. They may originate from customers, suppliers, internal processes, or regulatory bodies.

Types of Inputs

- Material inputs: Raw materials, components
- Information inputs: Customer orders, specifications, drawings
- Human inputs: Skills, competencies, approvals
- System inputs: Data from IT systems
- Regulatory inputs: Legal requirements, standards

Inputs must be:

- Clearly defined
- Verified for adequacy
- Available at the right time

Poor-quality inputs inevitably lead to poor outputs.

4. Outputs

Outputs are the results produced by a process. They are delivered to internal or external customers.

Types of Outputs

- Finished products
- Services delivered
- Reports or documents
- Decisions or approvals
- Updated records

Output Requirements

Outputs should meet:

- Customer requirements
- Statutory and regulatory obligations
- Quality, cost, and delivery expectations

Link to Customer Satisfaction

Outputs are the primary touchpoint between the organisation and its stakeholders.

5. Conditions (Operating and Control Conditions)

Conditions are the circumstances, constraints, and rules under which a process operates.

Types of Conditions

- **Environmental:** Temperature, cleanliness, safety conditions
- **Operational:** Capacity limits, shift patterns
- **Regulatory:** Legal compliance, industry standards
- **Organisational:** Policies, procedures, approvals
- **Risk-related:** Controls to prevent errors or failures

Defining conditions ensures:

- Repeatability
- Compliance
- Risk mitigation
- Stability of process outcomes

6. Resources

Resources are the assets required to perform a process effectively.

Categories of Resources

- **Human resources:** Trained personnel, competencies
- **Infrastructure:** Buildings, machines, tools
- **Equipment:** Measuring instruments, IT hardware
- **Software and systems:** ERP, CRM, design tools
- **Financial resources:** Budgets, cost limits
- **Knowledge resources:** Procedures, manuals, organisational know-how

Resource Management

Resources must be:

- Adequate
- Available
- Maintained
- Periodically reviewed for suitability

7. Criteria for Performance Measurement

Performance criteria define how well a process is performing and whether it achieves intended results.

Key Performance Indicators (KPIs)

Common metrics include:

- **Quality:** Defect rate, rework percentage
- **Delivery:** Cycle time, on-time delivery

- **Cost:** Cost per unit, waste reduction
- **Productivity:** Output per person/hour
- **Compliance:** Audit findings, deviations
- **Customer satisfaction:** Complaints, feedback scores

SMART Criteria

Performance measures should be:

- **Specific**
- **Measurable**
- **Achievable**
- **Relevant**
- **Time-bound**

Role in Continuous Improvement

Measurement enables:

- Trend analysis
- Root cause identification
- Data-driven decision-making
- PDCA and risk-based improvement

8. Integrated View of a Process

A well-defined process answers the following questions clearly:

Element	Key Question
Super process	How does this support business strategy?
Sub process	What specific activity is performed?
Inputs	What is required to start the process?
Outputs	What is delivered and to whom?
Conditions	Under what rules and constraints?
Resources	With what assets and competencies?
Performance criteria	How do we know it is working?

Defining processes in this structured manner transforms an organisation from a collection of activities into a controlled, measurable, and continuously improving system. Clear identification of super processes, sub processes, inputs, outputs, conditions, resources, and performance criteria is the foundation for quality management, risk control, operational excellence, and sustainable growth.

Standardization and Scalability

Repeatable processes reduce errors, improve speed, and enable delegation. Start by:

- Mapping workflows (e.g., order fulfilment, onboarding)
- Identifying bottlenecks and redundancies
- Creating Standard Operating Procedures (SOPs)

CREATING PROCESS MAPS

A process map is a visual representation of how work flows through an organisation. It shows:

- What activities are performed
- In what sequence
- By whom
- Using what resources
- Under what conditions
- With what expected outputs and performance criteria

Process maps convert written process definitions into operational clarity, making them essential for:

- ISO 9001 compliance
- Risk-based management

- Training and handover
- Continuous improvement

2. Step 1: Define the Process Boundary (Start and End)

Clear boundaries prevent:

- Overlapping responsibilities
- Missing interfaces
- Gaps between departments

Actions

- Identify the trigger (input that starts the process)
- Identify the final output (deliverable to a customer or next process)

Example

Process: Order to Cash

- **Start:** Receipt of customer order
- **End:** Receipt of customer payment

This defines the scope of the process map.

3. Step 2: Identify the Super Process and Sub Processes

Super Process Identification

Ask:

- Does this process directly create customer value?
- Does it cut across multiple functions?

Break into Sub Processes

Decompose the super process into logical, manageable steps.

Example Breakdown

Level	Process
Super Process	Order to Cash
Sub Process 1	Order Review & Confirmation
Sub Process 2	Production Planning

Sub Process 3	Manufacturing / Service Delivery
Sub Process 4	Dispatch
Sub Process 5	Invoicing & Collection

Each sub process will later become a lane or block in the process map.

4. Step 3: Identify Inputs and Outputs for Each Sub Process

Technique: SIPOC at Sub-Process Level

Before drawing the map, list:

- Supplier
- Input
- Process
- Output
- Customer

Example

Sub Process	Inputs	Outputs
Order Review	Customer PO, specs	Confirmed order
Planning	Confirmed order	Production schedule
Dispatch	Finished goods	Delivered product

Mapping Rule

Every activity box in the map must have:

- At least **one input**
- At least **one output**

5. Step 4: Sequence the Activities (Flow Logic)

Define Activity Flow

Arrange activities in:

- Chronological order
- Logical dependency order

Use Standard Flowchart Symbols

- **Oval:** Start / End
- **Rectangle:** Activity
- **Diamond:** Decision
- **Arrow:** Flow direction

Example Logic

Order Received → Order Reviewed → Credit Approved? → If Yes → Plan Production If No seek Clarification / Reject

The Yes/No decision points are critical—they often represent **risk and control points**.

6. Step 5: Assign Ownership and Resources (Swimlane Mapping)

Swimlane Concept

Swim lanes show **who does what**, preventing role confusion.

Typical Swim lanes

- Sales
- Planning
- Production
- Quality
- Finance

Identify Resources per Activity

For each activity, document:

Human Resources

- Role (not individual names)
- Competency required

Infrastructure & Tools

- Machines
- Software

- Measuring instruments

7. Step 6: Identify Conditions and Controls

Conditions define how the activity must be performed.

Typical Conditions to Map

- Regulatory requirements
- Work instructions
- Environmental limits
- Safety rules
- Approval levels

Control Points

Mark where:

- Verification occurs
- Approval is required
- Errors are likely

Example

- "Quality inspection as per Inspection Plan QP-04"
- "Dispatch only after QA release"

These become control nodes in the process map.

8. Step 7: Identify Risks and Failure Points (Optional but Recommended)

At each decision or hand-off, ask:

- What can go wrong?
- What is the impact?
- What control exists?

Example Risks

- Wrong order entry
- Capacity overload
- Late delivery

- Billing errors

Mark these as:

- Notes on the map
- Linked risk registers

This directly supports ISO 9001 risk-based thinking.

9. Step 8: Define Performance Criteria and Metrics

Linking Metrics to Process Steps

Metrics should be attached to specific sub processes, not just the overall process.

Typical Metrics by Category

Process Step	Metric	Purpose
Order Review	Order accuracy %	Quality
Planning	Schedule adherence	Delivery
Production	First pass yield	Efficiency
Dispatch	On-time delivery	Customer satisfaction
Billing	Invoice accuracy	Financial control

Metric Characteristics

Metrics must be:

- Clearly defined
- Measurable
- Reviewable at defined intervals

10. Step 9: Validate the Process Map

Validation Questions

- Are all inputs and outputs captured?
- Are responsibilities clear?
- Are risks addressed?
- Are metrics measurable with available data?

Stakeholder Review

Review the map with:

- Process owners
- Operators
- Quality team

11. Step 10: Maintain and Improve the Process Map

When to Update

- Process change
- Audit findings
- Customer complaints
- Performance deviations

Use in PDCA

- **Plan:** Map the process
- **Do:** Execute as mapped
- **Check:** Measure performance
- **Act:** Improve and update map

12. Integrated Example: What a Complete Process Map Captures

A good process map explicitly shows:

Element	Visible in Map
Super & sub processes	High-level structure
Inputs & outputs	Entry/exit points
Resources	Swim lanes & notes
Conditions	Controls & references
Metrics	KPI callouts
Risks	Decision points

Creating process maps is not merely a documentation exercise—it is a management discipline. By systematically identifying process boundaries, sequencing activities, assigning resources, defining conditions, and linking measurable criteria, organizations gain visibility, control, and improvement capability